From Listeners to Creators

Bridging Musical Inequality through Generative AI Music Tools
A Practical Study Based on the ADDIE Model in Junior High School Education

Yeh, Jung-Chen · Southern Taiwan University of Science and Technology
zoeyeh0822@gmail.com

AI-Driven Educational Scaling & Equity
Oral Presentation 1, Room B
11:40–12:00 | Paper ID: 24

Keywords:
Generative AI · Music Education · Cognitive Scaffolding · ADDIE Model · Educational Equity · UDL · SDG 4 · SDG 10

18 Classes — Grades 7–8

496 Student Creators — Junior High School

505 Original Compositions — May–June 2025

7+ Languages — Mandarin, English, Japanese, Korean, Vietnamese, Arabic…

● Background

Music is universally celebrated, yet composition has long remained the privilege of those with formal music theory training. Generative AI platforms such as SUNO AI have significantly lowered the technical threshold, shifting the creative barrier from technical mastery to conceptual expression.

In junior high school settings, however, students with typing difficulties or limited language proficiency frequently struggle to translate internal emotions into precise AI prompts — a challenge this study terms the "Conceptual Materialization Dilemma." Without deliberate pedagogical scaffolding, GenAI tools risk reproducing rather than dismantling existing inequalities in aesthetic education.

● Research Objectives

This study aimed to design, implement, and evaluate a three-tiered cognitive scaffolding system enabling students with no formal music training to compose original multilingual songs using SUNO AI. Grounded in the ADDIE model, the study sought to verify:

- Whether Human-AI collaboration can empower diverse learners to transcend linguistic barriers
- Whether creative agency can be restored to non-specialist student composers
- Whether technological equity can be actualized in a public junior high context

● Theoretical Foundations

ADDIE Model
Systematic instructional design:
Analysis → Design → Development →Implementation → Evaluation

Scaffolding Theory & ZPD
Temporary support calibrated to each learner's zone of proximal development (Vygotsky)

Cognitive Load Theory
Reduced extraneous load →
freed cognitive resources for creative investment (Sweller et al., 1998)

Flow Theory
Autotelic, deeply absorbing engagement;
Flow directly predicts music learning outcomes (Csikszentmihalyi, 1990)

Self-Determination Theory
Autonomy + Competence + Relatedness →
intrinsic motivation emerges and endures (Deci & Ryan, 2000)

UDL · SDG 4 · SDG 10
Universal Design for Learning; Quality Education and Reduced Inequalities as guiding values

● Research Context

Setting:	Public junior high school, Taiwan
Participants:	496 students, Grades 7–8, 18 classes
Duration:	May–June 2025, Semester 2, AY 2024–2025
Curriculum:	Cross-disciplinary: Technology × Arts
Prior Experience:	Same cohort had GenAI scaffolding experience in preceding semester

[Fig. 1: Students in Classroom During Six-Week Module]

● Methods & Implementation

A practitioner-research design was adopted across 18 heterogeneous classes (N = 496). The six-week cross-disciplinary module integrated Taiwan's Technology and Arts curriculum domains. Student outputs were archived in 18 class SUNO playlists and a Padlet digital gallery.

● Six-Week Implementation Timeline

Wks 1–2	**Exploration & Foundation** Song structure, 11 genres, rhyming, SUNO registration
Wks 3–4	**Scaffolded Creation** Theme selection → Prompt Website → Custom GPT → SUNO generation
Wks 5–6	**Refinement & Exhibition** Error correction, advanced features (Cover/Extend/Tempo), Padlet gallery & class playlists

● Three-Tiered Cognitive Scaffolding System

I have an idea → Tier 1 → Tier 2 → Tier 3 → I have a finished song

Tier 1 · Thematic Framework

9 major themes · 180 creative starting points calibrated to adolescent emotional life.

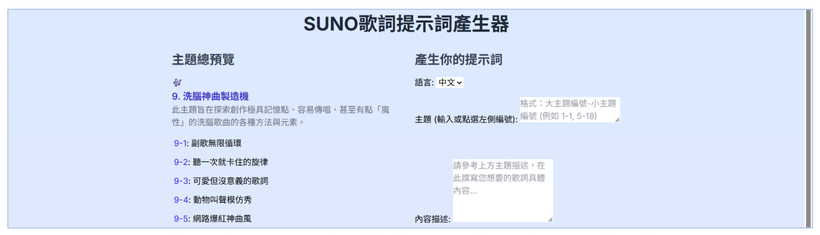
Students choose a sub-topic as an immediate emotional anchor — eliminates ideation paralysis entirely.

1. 心動的秘密信號	4. 載滿回憶的行囊	9. 洗腦神曲製造機
此主題旨在描寫青春期初次對人產生好感、暗戀時那種憧憬、害羞、酸甜又充滿秘密的心情與觀察。	此主題旨在透過回憶校園點滴、感謝師友、抒發離情並展望未來，創作適合在畢業典禮演唱或分享的歌曲。	此主題旨在探索創作極具記憶點、容易傳唱、甚至有點「魔性」的洗腦歌曲的各種方法與元素。
1-1: 看見你就會心跳加速	4-1: 教室最後的鐘聲	9-1: 副歌無限循環
1-2: 偷偷看你的側臉	4-2: 寫滿祝福的紀念冊	9-2: 聽一次就卡住的旋律
1-3: 不敢說出口的「喜歡」	4-3: 走過無數次的長廊	9-3: 可愛但沒意義的歌詞
1-4: 為什麼在意你的每個小動作？	4-4: 藏在心底的感謝	9-4: 動物叫聲模仿秀
1-5: 收到你訊息時的開心	4-5: 那年夏天的約定	9-5: 網路爆紅神曲風
1-6: 擦肩而過的瞬間	4-6: 校門口的道別與期許	9-6: 兒童節目主題曲Fu
1-7: 希望你能注意到我	4-7: 收進行囊的青春碎片	9-7: 超級洗腦廣告歌

[Fig. 2: 9x20 Thematic Framework Matrix]

Tier 2 · Prompt Scaffolding Website

Google Sites platform: language selector (CN / EN / KO / JA), genre tags, phonetic rhyme-scheme selector, vocal arrangement dropdown. One-click prompt copy auto-generates a complete structured prompt. No English prompt typing required; mobile-accessible.



[Fig. 3: Prompt Scaffolding Website Screenshot]

Tier 3 · Custom GPT: SUNO Lyrics Writer

Accepts Mandarin input; outputs fully structured lyrics with structural markers [Verse/Chorus/Bridge] in any target language. Automatic Mandarin translation + romanized phonetics appended. Adds only structure — never alters student creative content.

[Fig. 4: SUNO Lyrics Writer — Custom GPT Interface]

Languages Bridged: Mandarin · English · Korean · Japanese · Vietnamese · Arabic · and more

● Results & Impact

505	496	18
Original Compositions	Student Creators	Class Playlists

Compositions spanned Mandarin, English, Korean, Japanese, Vietnamese, Arabic, and additional languages — demonstrating that the scaffolding system effectively bridges virtually any linguistic divide. Qualitative observation across all 18 classes revealed sustained collective Flow states during Weeks 3–6, with behavioral disruptions virtually absent.

3 students composed entirely self-authored lyrics; the Custom GPT added only structural markers — evidence of genuine creative ownership cultivated by the scaffolding.

Numerous students continued creating autonomously after the formal curriculum concluded — intrinsic motivation genuinely internalized.

Graduation song projects, entirely student-initiated, were recognized by faculty as artistically meritorious.

[Fig. 5: Padlet Digital Gallery — 496 Student Works]

[Fig. 6: SUNO Class Playlists (18 classes, 25–30 songs each)]

● Discussion & Conclusion

This study demonstrates that GenAI tools, paired with intentionally designed cognitive scaffolding, can democratize aesthetic creation in junior high contexts. The educational divide in GenAI music creation resides not in hardware access but in cognitive and linguistic resources — addressable through Universal Design for Learning principles.

The ADDIE-grounded three-tiered scaffolding framework offers educators a replicable, tool-agnostic blueprint for large-scale AI integration. Prior student AI literacy meaningfully accelerates adoption.

● Three Key Implications

① Democratizing Aesthetic Creation
Flow states can be systematically engineered — novices enter them too. The challenge-skill balance is engineerable.

② Educational Equity (SDG 4 & 10)
Divide resides not in hardware but in cognitive & linguistic resources. UDL principles address this gap directly.

③ Replicable, Tool-Agnostic Blueprint
Google Sites + Custom GPT: replicable, requiring ~2 weeks design investment. ADDIE is transferable across contexts.

● Future Research Directions

Future research should incorporate validated instruments measuring flow states, creative self-efficacy, and longitudinal autonomous learning behaviors. Exploration of cross-school replication and adaptation for other STEM-Arts cross-disciplinary contexts is recommended.

Email
Spotify
Prompt Scaffolding Website
SUNO Lyrics Writer
Padlet Digital Gallery
Teaching Materials